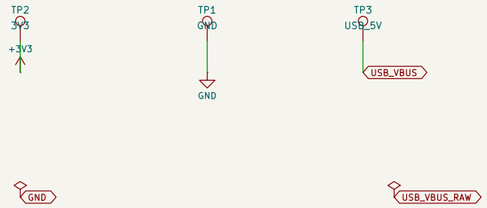
01 POWER AND MCU SUPPLIES

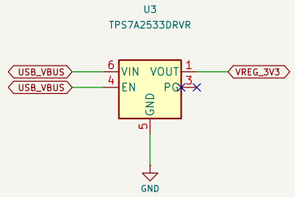
Global signal labels connect sheets. Pin numbers refer to the physical LQFP-144 package.

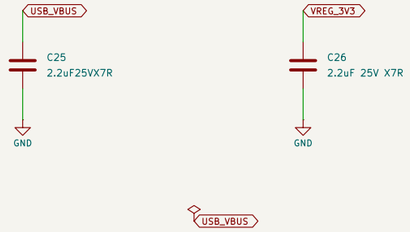
USB INPUT ISOLATION



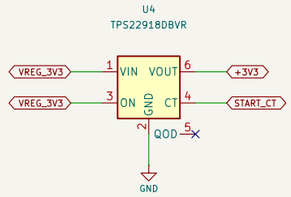


3.3 V REGULATOR



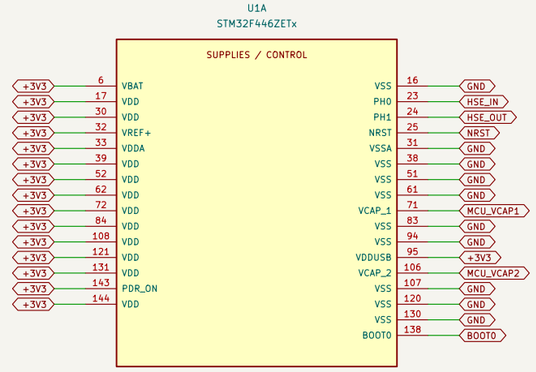


CONTROLLED RAIL STARTUP

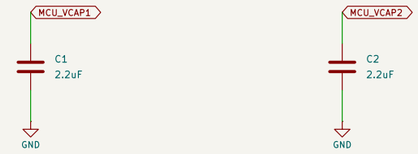


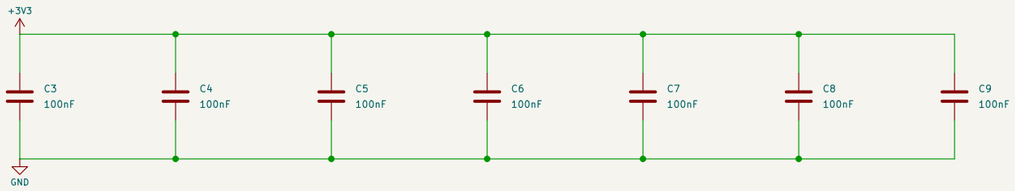


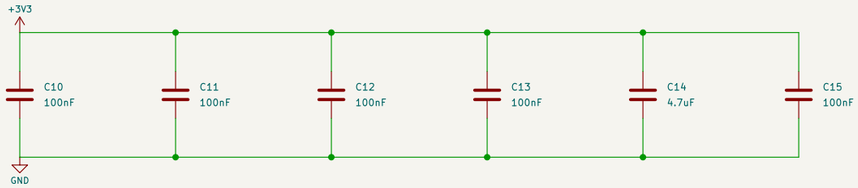
MCU POWER / CONTROL

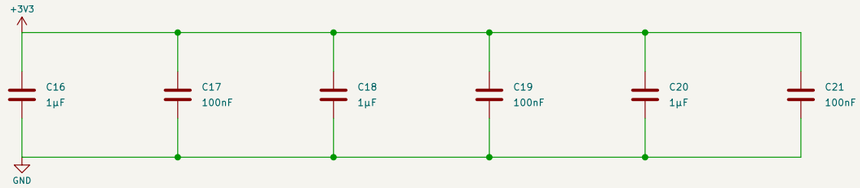


MCU BYPASS CAPACITORS









Power budget: 250 mA total is conditional; thermal and load testing are pending.
Do not power motors or inject another 3.3 V source into this rail.

USB, debug, reset and clock

File: usb_debug_clock.kicad_sch

MCU ports and peripheral interfaces

File: mcu_interfaces.kicad_sch

GPIO expansion

File: gpio_expansion.kicad_sch

Presentation refactor — hardware validation pending

Sheet: /
File: stm32.kicad_sch

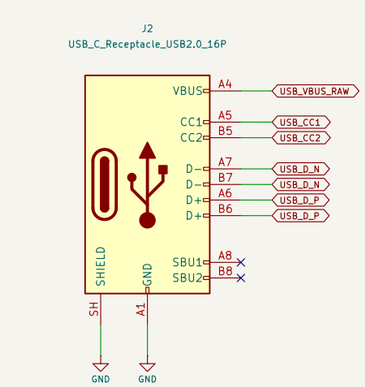
Title: STM32F446 | Power and MCU supplies

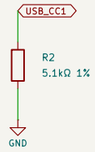
Size: A3	Date: 2026-10-16	**Rev: Prototype**
KiCad E.D.A. 10.0.6		Id: 1/4

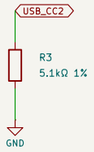
02 USB, DEBUG, RESET AND CLOCK

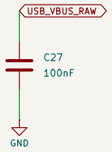
Global signal labels connect sheets. Pin numbers refer to the physical LQFP-144 package.

USB DEVICE CONNECTOR



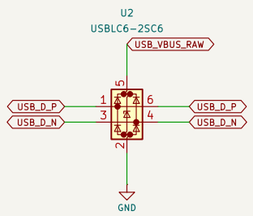




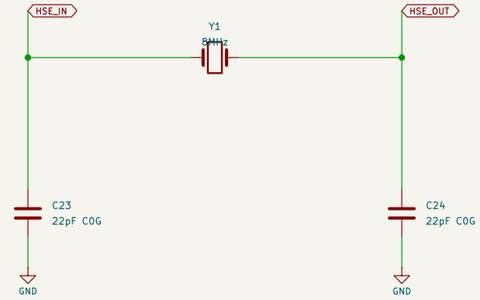




DATA-LINE PROTECTION



8 MHz HSE CRYSTAL

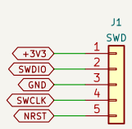


RESET



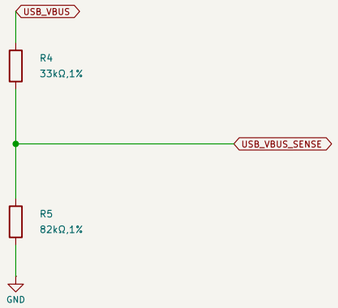


EXTERNAL SWD



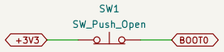
J1: 1=3.3 V reference, 2=SWDIO, 3=GND, 4=SWCLK, 5=NRST.
Custom pinout: map the probe adapter explicitly.

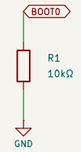
VBUS SENSING

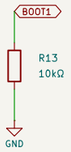


PA9 sensing configuration is a firmware requirement.
This divider is not a general-purpose 3.3 V logic input.

BOOT CONFIGURATION







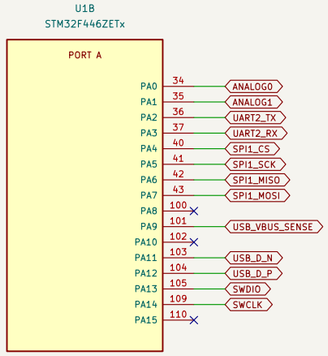
Presentation refactor — hardware validation pending

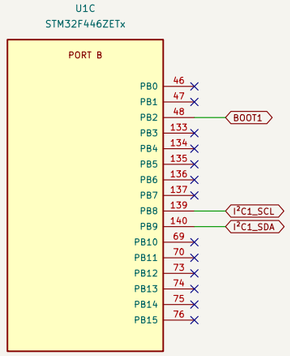
Sheet: /USB, debug, reset and clock/ File: usb_debug_clock.kicad_sch		
Title: STM32F446 | USB, debug, reset and clock		
Size: A3	Date: 2026-10-16	**Rev: Prototype**
KiCad E.D.A. 10.0.6		Id: 2/4

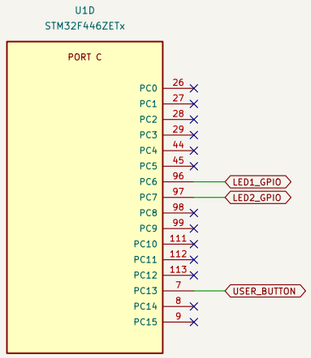
03 MCU PORTS AND PERIPHERAL INTERFACES

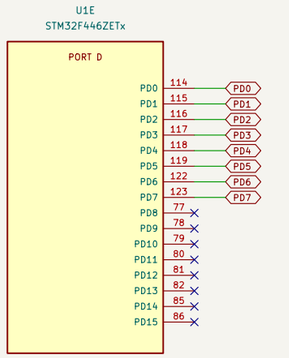
Global signal labels connect sheets. Pin numbers refer to the physical LQFP-144 package.

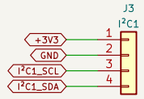
MCU GPIO PORTS

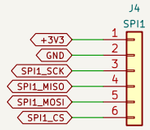


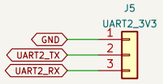


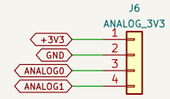


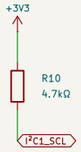




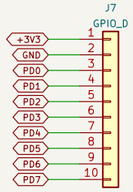






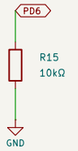






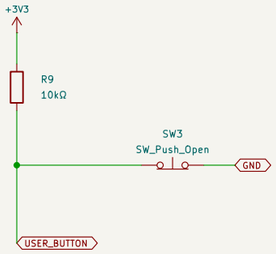


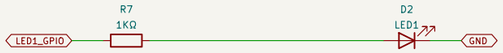




CAN1 on J7: pin 3=PD0/RX, pin 4=PD1/TX (AF9).
External CAN transceiver and termination required: link testing pending.

USER INDICATORS / BUTTON







Presentation refactor — hardware validation pending

Sheet: /MCU ports and peripheral interfaces/
File: mcu_interfaces.kicad_sch

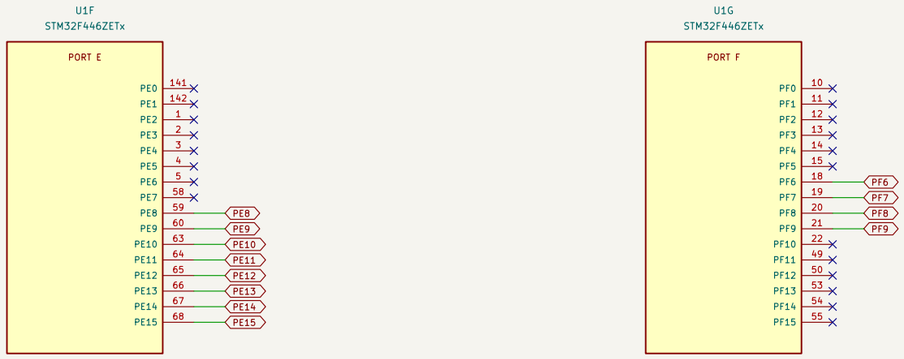
Title: STM32F446 | MCU ports and peripheral interfaces

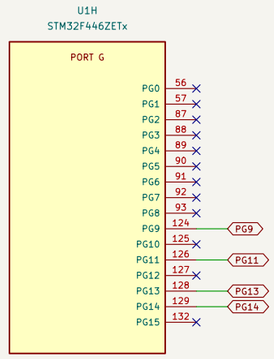
Size: A3	Date: 2026-10-16	**Rev: Prototype**
KiCad E.D.A. 10.0.6		Id: 3/4

04 GPIO EXPANSION

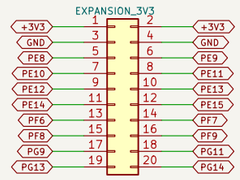
Global signal labels connect sheets. Pin numbers refer to the physical LQFP-144 package.

EXPANSION GPIO PORTS





20-PIN EXPANSION HEADER



POWER INDICATOR



J8 groups selected PE / PF / PG pins with 3.3 V and GND.
All expansion modules share the board power budget.
Unused MCU pins are explicitly marked no-connect.

Presentation refactor — hardware validation pending

Sheet: /GPIO expansion/
File: gpio_expansion.kicad_sch

Title: STM32F446 | GPIO expansion

Size: A3	Date: 2026-10-16	**Rev: Prototype**
KiCad E.D.A. 10.0.6		Id: 4/4